Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side *(e.g., React, Angular; HTML/CSS)*	React.js (with Vite), HTML5, CSS3, JavaScript (ES6+)	*React.js provides a fast and responsive UI with reusable components. Vite offers a faster build and development experience. HTML5, CSS3, and JavaScript ensure modern, responsive, and cross-browser compatible UI.*
2	Backend / Server-Side *(e.g., Node.js, Python; Spring Boot)*	*Python, FastAPI*	*FastAPI is a modern, high-performance Python web framework ideal for building RESTful APIs with automatic validation and documentation.*
3	Database *(e.g., MySQL, PostgreSQL; MongoDB)*	*SQLite*	*SQLite is lightweight, serverless, and easy to set up. It is ideal for this application's data storage needs and ensures portability.*
4	ORM / Database Toolkit *(e.g., SQLAlchemy, Prisma)*	*SQLAlchemy ORM*	*SQLAlchemy ORM provides powerful and flexible database interaction with Python, making it easier to manage models and queries.*
5	AI / ML Services *(e.g., OpenAI API, Google Gemini API)*	*Google Gemini AI API*	*Gemini AI API is used for financial analysis, settlement prediction, negotiation strategies, and AI letter generation.*
6	API Communication *(e.g., REST, GraphQL)*	*RESTful APIs*	*RESTful APIs are used for communication between frontend and backend, ensuring scalability and easy integration.*
7	Development Tools *(e.g., VS Code, Git)*	*VS Code, Git, GitHub*	*VS Code is used as the IDE. Git and GitHub are used for version control, collaboration, and code management.*
8	Deployment *(e.g., Vercel, Render, AWS, Docker)*	*Render (Backend), Vercel (Frontend)*	*Render is used to deploy the FastAPI backend, and Vercel is used for deploying the React frontend for fast and reliable hosting.*